A Comparative Flood Risk Assessment of Lagos (Nigeria) and Durban (South Africa)

Integrating Climate Science, Remote Sensing, Spatial Analysis and Financial Scenario Modelling

Author: *Sihle Nala*

Executive Summary

Urban flooding has become one of the most damaging natural hazards affecting rapidly growing cities across Africa. Increasing urbanisation, expanding impervious surfaces, ageing drainage infrastructure, and more intense rainfall events have increased both the frequency of flooding and its economic consequences. Coastal cities are particularly vulnerable because flood hazards interact with dense populations, critical infrastructure, and expanding economic activity.

This study presents a comparative assessment of flood risk in Lagos, Nigeria, and Durban, South Africa. The analysis covers historical observations from 1990 to 2023 and develops future flood damage scenarios for the period 2025 to 2050. An integrated analytical framework was adopted to investigate the physical, climatic, spatial, and economic dimensions of flood risk. Climate trends were evaluated using ERA5 reanalysis data, historical flood impacts were examined using the EM-DAT disaster database, population trends were obtained from World Bank indicators, and spatial analyses incorporated the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model together with Global Human Settlement Layer (GHSL) built-up data. These datasets were analysed in Python using climate trend analysis, flood disaster analysis, digital elevation modelling, built-up land assessment, composite spatial flood risk modelling, ordinary least squares (OLS) regression, financial scenario modelling, and cost-benefit analysis.

The study was guided by four research questions. First, it investigated the factors responsible for flooding in Lagos and Durban. Second, it examined whether flood risk has increased over time. Third, it assessed the current spatial distribution of flood risk within each city. Finally, it evaluated which flood mitigation strategies provide the greatest economic return.

The results demonstrate that flood risk in both cities is influenced by a combination of climatic and urban factors rather than changes in annual rainfall alone. Although long-term precipitation trends remain relatively stable, flood impacts continue to increase because urban expansion, land-use change, ageing infrastructure, and more intense rainfall events have increased exposure and vulnerability. The findings indicate that Lagos is primarily characterised by structural urban vulnerability. Low-lying terrain, extensive impervious surfaces, dense development, and inadequate drainage systems create widespread flood susceptibility across the city. Durban exhibits a different flood regime, where steep slopes and confined river catchments accelerate runoff generation and contribute to localised flash flooding during intense rainfall events.

Spatial flood risk modelling reinforced these differences. Lagos recorded a higher composite flood risk and more than twice the proportion of land classified as high risk compared with

Durban. Although Durban exhibited a slightly lower average flood risk, steep terrain created concentrated areas of elevated hazard that remain highly vulnerable during extreme rainfall events.

Regression modelling showed that climatic variables alone cannot fully explain observed flood damages. The statistical models consistently underestimated historical economic losses because they do not capture local drainage deficiencies, informal settlement expansion, infrastructure quality, or other structural drivers of flood vulnerability. This highlights the importance of integrating statistical modelling with spatial analysis and historical disaster records when assessing urban flood risk.

Future scenario modelling projected continued increases in flood damages under a baseline scenario with no additional mitigation measures. The financial evaluation demonstrated that investment in flood mitigation can substantially reduce future economic losses. In Lagos, drainage rehabilitation produced the largest reduction in cumulative flood damages and generated a positive net present value despite requiring substantial initial investment. Early warning systems achieved the highest benefit-cost ratio because relatively modest implementation costs generated considerable economic benefits and earlier financial payback. In Durban, both retention measures combined with vegetation restoration and early warning systems reduced projected flood losses. However, the financial returns were smaller than those observed in Lagos because of lower projected damages and the nature of the interventions evaluated. Even so, early warning systems remained economically viable because of their comparatively low implementation costs.

Overall, the study demonstrates that flood mitigation strategies should reflect the dominant flood mechanisms within each city. Large-scale drainage improvements are particularly effective in Lagos, where flooding is strongly associated with inadequate urban infrastructure. Durban benefits more from integrated catchment management approaches that combine runoff retention, vegetation restoration, and improved preparedness. Across both cities, early warning systems emerged as the most cost-effective intervention because they provide substantial reductions in flood impacts while requiring relatively low capital investment.

The findings illustrate the value of integrating climate data, disaster records, remote sensing, spatial modelling, statistical analysis, and economic evaluation within a single analytical framework. This approach provides evidence that can support climate adaptation planning, infrastructure investment, and urban flood risk management in rapidly growing African cities. The framework developed in this study is readily transferable to other urban areas facing similar challenges and provides a practical basis for prioritising flood mitigation investments under increasing climate and urbanisation pressures.

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1. Introduction

Flooding ranks among the most expensive natural disasters globally, and its effects are becoming more severe in many rapidly urbanising regions. African cities face unique challenges, as urban growth often outpaces the development of essential infrastructure, leaving increasing numbers of people vulnerable to flood risks. Climate change exacerbates these issues, leading to rising temperatures, altered rainfall patterns, and an uptick in extreme weather events.

Lagos and Durban, two economically significant coastal cities, frequently experience flooding but differ greatly in their physical geography and urban development trends. Lagos lies on a low-lying coastal plain filled with lagoons, wetlands, and vast reclaimed areas, whereas Durban is positioned in steep coastal catchments where quick runoff from elevated land often causes flash floods.

Grasping these differences is crucial for creating effective adaptation strategies. While some solutions may work well in one city, they might not be as beneficial in another. This underscores the need for city-specific assessments to guide evidence-based planning.

Most prior flood research has either concentrated on climatic patterns or engineering solutions. Fewer studies have integrated climate data, spatial flood vulnerability, statistical damage modelling, and financial analysis within a cohesive framework. This research fills that gap by merging multiple datasets and analytical methods to explore both the physical and economic aspects of flood risk.

Additionally, the study illustrates how geospatial analysis can aid financial decision-making by correlating spatial flood risk with anticipated economic damages and assessing the costs and benefits of various adaptation measures.

1.1 Research Objectives

The study aims to compare flood risk between Lagos and Durban by integrating climate observations, topographic analysis, built environment characteristics and historical disaster records.

The specific objectives are to:

- investigate the major physical drivers of flooding in each city;
- determine whether flood occurrence and damages have increased over time;
- map the spatial distribution of flood risk using remotely sensed datasets;
- estimate baseline annual flood damages using regression models;
- evaluate future mitigation scenarios using financial cost-benefit analysis; and
- identify interventions that provide the greatest long-term economic return.

1.2 Research Questions

Table 1: Research questions addressed in the study and the corresponding analytical approaches used to investigate each question.

Research Question	**Analytical Approach**
Why is flooding occurring?	Literature review, climate analysis and spatial assessment
Has flooding increased over time?	Trend analysis using historical flood records
What is the spatial distribution of flood risk?	DEM analysis, built-up density mapping and composite flood risk modelling
Which mitigation strategies provide the greatest economic benefit?	Financial scenario modelling and cost-benefit analysis

1.3 Data Sources

Five primary datasets were integrated throughout the analysis. Together, these datasets provide the climatic, demographic, environmental and socioeconomic information required to investigate flood risk from multiple perspectives.

Table 2: Primary datasets used in the study, including their sources and analytical purposes.

Dataset	**Purpose**
ERA5 Reanalysis	Annual precipitation and temperature
EM-DAT Disaster Database	Historical flood events and economic losses
World Bank Indicators	Population growth
SRTM Digital Elevation Model	Elevation and slope
GHSL Built-up Data	Urban development and impervious surface density

1.4 Analytical Framework

The analytical workflow followed five sequential stages:

1. Collection and preprocessing of climate, disaster and geospatial datasets.
2. Development of composite flood risk maps using elevation, slope and built-up density.
3. Estimation of baseline annual flood damages through regression modelling.
4. Projection of future flood damages under baseline and intervention scenarios (2025-2050).

5. Evaluation of intervention performance using Net Present Value (NPV), Benefit-Cost Ratio (BCR) and break-even analysis.

This integrated framework links environmental processes with economic decision-making, allowing flood mitigation options to be assessed not only for their ability to reduce hazard exposure but also for their long-term financial viability.

2. Why Is Flooding Happening?

2.1. Introduction

Flooding is rarely caused by a single factor. Instead, it results from interactions between climate, topography, land use, urban growth and drainage infrastructure. Although Lagos and Durban both experience recurrent flooding, the mechanisms responsible differ considerably.

To understand these differences, this study combines climate observations, terrain analysis, built-up surface mapping and published literature. The objective is to identify the dominant flood drivers in each city before examining their economic consequences.

The analysis demonstrates that climate change alone cannot explain the observed flood impacts. Human modification of the landscape has become equally important in determining flood severity.

2.2. Climatic Drivers

Climate plays a significant role in flooding primarily through factors like the intensity and duration of rainfall, as well as the moisture levels in the soil beforehand. Rising temperatures also have an impact on the hydrological cycle by increasing the amount of moisture in the atmosphere, which creates conditions that lead to more intense rainfall events.

Data on annual precipitation from the ERA5 reanalysis indicates that neither Lagos nor Durban has shown a consistent increase in total annual rainfall over recent years. Despite this, both cities are facing increasingly severe flood events.

This seeming contradiction highlights a crucial difference between the quantity of rainfall and its intensity. Instead of receiving much more rainfall annually, both cities are becoming more vulnerable to short, intense storms that can easily overwhelm urban drainage systems.

As temperatures rise, the atmospheric capacity to hold water increases, following the Clausius-Clapeyron relationship. This allows storms to produce heavier rainfall over shorter timeframes. As a result, flood frequency and damage may rise even if the overall annual rainfall remains fairly stable.

2.3. Lagos: A Structurally Vulnerable Coastal Megacity

Low Elevation

Lagos stands as one of Africa's lowest-lying megacities. Much of its vast metropolitan area is situated just a few meters above sea level and is encircled by lagoons, wetlands, and tidal creeks. This low elevation hampers natural drainage, making it difficult for runoff to flow away from developed regions. During high tides, drainage channels can become partially obstructed due to the backwater effects from both the Atlantic Ocean and Lagos Lagoon. As a result, even a moderate amount of rainfall can lead to widespread surface flooding.

Rapid Urbanisation

Lagos has witnessed an incredibly rapid pace of urban growth in recent decades. This expansion has led to the replacement of wetlands and green areas with roads, buildings, and paved surfaces. These impervious surfaces hinder water infiltration and contribute to increased runoff volume. Instead of being absorbed by the soil, rainfall quickly rushes over concrete and asphalt, gathering in low-lying areas or overwhelming drainage systems. Satellite-derived data from the Global Human Settlement Layer (GHSL) vividly highlights the concentration of dense urban development across much of metropolitan Lagos.

Drainage Infrastructure

The expansion of drainage infrastructure has not kept pace with the rapid growth of the population. Many drainage channels are currently facing several issues, including sediment buildup, blocked culverts, illegal dumping, lack of proper maintenance and inadequate hydraulic capacity. Moreover, informal settlements often don't have any engineered stormwater systems in place. Consequently, rainfall events that should be manageable often lead to significant urban flooding.

Human Exposure

Flood impacts are magnified by the high concentration of people, businesses, and transportation infrastructure in areas prone to flooding. Consequently, economic losses stem not only from the frequent occurrences of floods but also from the large number of individuals and valuable assets located in hazardous zones. Lagos is particularly susceptible to urban flooding due to its combination of low elevation, extensive impervious surfaces, and insufficient drainage systems.

2.4. Durban: Topography-Controlled Flooding

Durban stands in stark contrast to Lagos. While Lagos boasts extensive coastal plains, Durban is marked by steep hills, deeply carved valleys, and short river catchments. These unique physical features set the stage for conditions that can easily lead to flash flooding.

Steep Slopes

When rain falls on steep slopes, it quickly rushes downwards. The water has little time to soak into the ground before it reaches the valleys below. As a result, river levels can rise sharply during heavy rainfall. This rapid increase in water flow can lead to flash floods that may develop in just a few hours.

River Catchments

Several rivers flow through Durban before emptying into the Indian Ocean. Among these are the Umgeni River, the Umbilo River, and the Isipingo River. When heavy rainfall occurs in the upstream catchments, it can lead to swift flooding downstream. As a result, urban development situated in the valley floors becomes vulnerable to both river flooding and local surface runoff.

Urban Expansion

Population growth has increasingly pushed residential development onto steep hillsides and flood-prone areas. Many informal settlements are situated in locations that are highly vulnerable to flooding and landslides. As more people are exposed to these risks, the impact of the same rainfall event is felt by a larger population, leading to greater economic losses.

Climate Sensitivity

In contrast to Lagos, statistical modelling indicates that rainfall plays a significant role in driving flood damage in Durban. The area's topography enables heavy rainfall to quickly turn into flood runoff. As a result, variations in storm intensity have a more immediate impact on flooding consequences.

2.5. Comparison Between Lagos and Durban

Although both cities experience flooding, the dominant drivers differ substantially. The comparison illustrates that flood management strategies should reflect local environmental conditions rather than adopting a single approach for all cities.

Table 3: Comparison of the dominant physical and urban drivers of flood risk in Lagos and Durban.

Flood Driver	**Lagos**	**Durban**
Low elevation	Very High	Moderate
Steep slopes	Low	Very High
Drainage limitations	Very High	Moderate
Urbanisation	Very High	High
Flash flooding	Moderate	Very High
Coastal influence	High	Moderate

2.6. Key Findings

The analysis highlights various key mechanisms driving flooding in different areas. In Lagos, urban structure and limited drainage are the main factors contributing to flood vulnerability. Meanwhile, in Durban, the terrain and swift runoff generation play a crucial role in flood behaviour. Climate change exacerbates these issues in both cities by intensifying rainfall, though the ways damage occurs differ between them. These insights lay the groundwork for the spatial risk assessment that will be discussed later in the report.

3. Has Flooding Increased Over Time?

3.1. Introduction

The second research question explores whether the frequency and severity of flooding have risen over time. To uncover long-term trends, researchers analysed historical flood records in conjunction with climate data. Understanding if flood risk is on the upswing is crucial for effective infrastructure planning, setting insurance rates, and adapting to climate change.

3.2. Historical Flood Records

Flood events documented in the EM-DAT database show that both Lagos and Durban have faced repeated flood disasters over the study period. Although the impact varies significantly from year to year, the overall trend points to increasing severity. In recent decades, both cities have witnessed some of their largest recorded flood disasters. These incidents have resulted in substantial economic losses, widespread damage to infrastructure, and the displacement of many communities.

3.3. Rainfall Trends

Data from ERA5 indicates that annual rainfall totals can vary significantly from one year to the next. However, there is no clear trend showing an overall increase in total annual rainfall. This observation aligns with findings from numerous recent climate studies, which suggest that the risk of flooding is increasingly influenced by rainfall intensity rather than simply the total amount of rainfall received each year.

3.4. Temperature Trends

Annual mean temperatures show a steady upward trend over the course of the study period. As warmer atmospheric conditions lead to increased evaporation and more moisture being stored in the atmosphere, these changes create favourable conditions for heavier rainfall events. While overall annual rainfall amounts remain fairly stable, the frequency of extreme rainfall events is rising, significantly heightening the risk of severe flooding.

3.5. Urban Growth

Population growth has significantly increased flood exposure. Urban expansion has produced larger impervious surfaces, reduced infiltration, increased runoff and greater numbers of people exposed to flooding. Consequently, similar rainfall events now produce substantially larger economic losses than in previous decades.

3.6. Interpretation

The combined evidence suggests that increasing flood damage cannot be attributed solely to climate. Instead, three interacting processes may explain the observed trends. These processes include increasing rainfall intensity, rapid urbanisation and expanding exposure of people and infrastructure. Flooding therefore reflects both environmental change and urban development.

3.7. Implications

The results indicate that future flood risk is likely to continue increasing unless urban planning and flood mitigation improve. Traditional drainage systems designed using historical rainfall statistics may become progressively less effective under changing climatic conditions. Planning based solely on historical flood records is therefore likely to underestimate future risk.

3.8. Transition to Spatial Analysis

The previous sections explain why flooding occurs and why impacts have increased. The next stage of the analysis investigates where flood risk is concentrated within each city. Using Digital Elevation Models (DEM), slope analysis and satellite-derived built-up density, a composite flood risk index is developed to identify areas most susceptible to flooding.

4. Spatial Flood Risk Assessment and Damage Modelling

4.1. Introduction

Understanding why flooding occurs is only part of flood risk assessment. Decision-makers also need to know where flood risk is concentrated and how that risk translates into economic losses. Spatial analysis provides this missing dimension by identifying areas that are most susceptible to flooding based on terrain characteristics and urban development.

A composite flood risk model was developed by integrating three spatial variables: Digital Elevation Model (DEM), terrain slope and Built-up surface density. Each variable represents an important component of flood susceptibility. Elevation determines where water naturally accumulates, slope influences runoff behaviour, and built-up density reflects the extent of impervious surfaces that reduce infiltration. The resulting composite flood risk maps provide a quantitative measure of flood susceptibility across Lagos and Durban and serve as the foundation for the financial scenario analysis presented later in this report.

4.2. Development of the Composite Flood Risk Index

Flood risk was estimated using raster-based spatial analysis in Python. All spatial datasets were projected onto a common coordinate reference system before analysis to ensure pixel-level alignment. Each raster was normalised to values between 0 and 1, allowing the variables to be combined into a single composite flood risk index. Higher values indicate greater flood susceptibility.

Table 4: Variables incorporated into the composite flood risk index, their data sources, and their influence on flood susceptibility.

Variable	Description	Influence on Flood Risk
Elevation	Digital Elevation Model (DEM)	Lower elevations generally experience greater flood susceptibility because water naturally accumulates in depressions and coastal plains.

Slope	Derived from DEM	Slope influences runoff behaviour. Flat terrain promotes water accumulation, whereas steep terrain promotes rapid runoff and flash flooding.
Built-up Density	GHSL Built Surface	Dense urban development increases impervious surfaces, reducing infiltration and increasing runoff volumes.

4.3. Flood Risk Results

The composite flood risk maps indicate substantial differences between Lagos and Durban. The results indicate that Lagos has both a higher average flood risk and a substantially larger proportion of land classified as high risk. Almost one-fifth of Lagos falls within the highest flood susceptibility category, compared with less than one-tenth of Durban. Although the difference in mean risk appears relatively small (0.454 versus 0.428), the difference in high-risk land area is considerable. This reflects the contrasting physical environments of the two cities.

Table 5: Summary statistics of the composite flood risk index for Lagos and Durban, including the mean flood risk and the proportion of land classified as high flood risk.

City	Mean Flood Risk	High-Risk Area (%)
Lagos	0.454	18.9
Durban	0.428	8.5

4.4. Spatial Characteristics of Lagos

The flood risk map for Lagos highlights extensive areas that are highly susceptible to flooding. The zones most at risk are predominantly found in coastal districts, lagoon edges, reclaimed lands and densely populated urban centers. These regions share three key features that heighten their vulnerability to flooding, which are very low elevation, dense urban development and limited capacity for water absorption.

Additionally, the vast network of lagoons and tidal waterways hampers drainage efficiency during heavy rain. Urban sprawl has transformed wetlands and natural flood storage areas into roads, housing, and commercial buildings, leading to swift surface runoff from rainfall. This spatial analysis underlines that Lagos faces a flood regime primarily driven by urbanization, where physical exposure and infrastructure constraints worsen the impact of climatic hazards.

4.5. Spatial Characteristics of Durban

Durban presents a more varied flood risk landscape. Instead of having broad areas of high risk, flood susceptibility is mainly found in specific locations, including river valleys, floodplains, coastal lowlands, and steep catchments. In contrast to Lagos, where low elevation tends to drive flood behaviour, the flood risks in Durban are significantly influenced by the terrain.

Steep slopes can quickly channel runoff into the lower-lying urban regions, leading to flash floods during heavy rainfall events. As a result, high-risk areas often occur where river systems cut through densely populated neighbourhoods. The relatively low percentage of high-risk land (8.5%) suggests that flood hazards are more spatially concentrated rather than evenly spread throughout the city.

4.6. Comparison Between the Two Cities

The spatial analysis highlights two distinct flood risk profiles.

Lagos

- Widespread urban flood susceptibility
- Extensive low-lying terrain
- High impervious surface coverage
- Large continuous high-risk zones

Durban

- Localised flood hotspots
- Strong topographic influence
- Flash flood behaviour
- Concentrated valley-floor exposure

Although both cities experience severe flooding, the underlying spatial processes differ considerably. These differences have important implications for flood management strategies. Large-scale drainage improvements are likely to have greater benefits in Lagos, whereas catchment management and runoff attenuation are more appropriate for Durban.

5. Regression Analysis and Damage Modelling

5.1. Introduction

Spatial flood risk identifies where flooding is likely to occur. Regression modelling extends this analysis by estimating the economic consequences of flooding. Statistical models were developed to estimate annual flood damages using climate and demographic variables. The resulting damage estimates provide the financial baseline required for future scenario modelling.

5.2. Baseline Flood Damage Estimates

The regression models produced the following baseline annual flood damage estimates. The predicted values represent expected annual damages generated by the statistical models. The observed values represent cumulative historical disaster losses derived from EM-DAT records.

Table 6: Comparison of regression-predicted annual flood damages and historically observed annual flood damages for Lagos and Durban.

City	Predicted Damage (USD million)	Historical Observed Damage (USD million)
Lagos	15.85	225.00
Durban	8.54	60.00

5.3. Interpretation of the Damage Estimates

At first glance, the difference between predicted and observed damages appears substantial. For example, the regression model estimates annual flood damage in Lagos at approximately **USD 15.85 million**, whereas historical observations indicate cumulative damages of approximately **USD 225 million**.

Similarly, Durban's predicted annual damage is **USD 8.54 million**, compared with historical reported damages of **USD 60 million**. This difference does not indicate poor model performance. Instead, it reflects differences in what the two measures represent. The regression model estimates the expected annual damage based on average climatic and demographic conditions. Historical observations include rare, high-impact disasters that contribute disproportionately to cumulative losses. Extreme flood events occur infrequently but account for a large proportion of total economic damage. Consequently, regression models generally underestimate the financial impact of catastrophic events while providing realistic estimates of expected annual losses.

5.4. Why the Models Underestimate Historical Damages

Several factors explain the observed differences. First, historical disaster databases include exceptional flood events that are difficult to predict statistically. Second, the regression models rely on a relatively limited number of annual observations. Third, important local variables such as drainage capacity, land-use planning, stormwater infrastructure and informal settlement density were not available as explanatory variables. These factors are particularly important in Lagos, where urban infrastructure plays a dominant role in determining flood severity. Consequently, the regression models should be interpreted as providing conservative baseline estimates rather than maximum potential losses.

5.5. Role of the Regression Models

Despite their limitations, the regression models remain valuable because they provide consistent baseline annual damages; comparable estimates between cities; quantitative inputs for future scenario modelling; and a framework for evaluating the economic benefits of flood mitigation interventions. These baseline estimates form the starting point for projecting future flood damages under different intervention scenarios.

5.6. Integrating Spatial and Financial Risk

The spatial flood maps and regression models complement one another. The flood maps identify where flooding is most likely to occur. The regression models estimate how much damage flooding is likely to cause. Together, they provide a comprehensive framework for assessing urban flood risk. Areas identified as having high spatial flood susceptibility are expected to contribute disproportionately to future economic losses. This integrated approach enables decision-makers to prioritise investments in locations where flood mitigation is likely to deliver the greatest economic benefit.

5.7. Key Findings

The analysis produces several important findings.

- Lagos has the highest overall flood susceptibility, with a **mean flood risk of 0.454**.
- Approximately **18.9%** of Lagos is classified as high-risk, more than double the proportion identified in Durban.
- Durban has a **mean flood risk of 0.428**, with **8.5%** of land classified as high-risk.
- Estimated baseline annual flood damages are **USD 15.85 million** for Lagos and **USD 8.54 million** for Durban.
- Historical disaster losses substantially exceed regression estimates because catastrophic flood events contribute disproportionately to long-term economic damages.
- Combining spatial flood risk analysis with statistical damage modelling provides a robust framework for evaluating future flood mitigation investments.

6. Financial Scenario Analysis and Flood Mitigation Strategies

6.1. Introduction

Understanding where flood risk occurs is only one component of effective decision-making. Governments, municipalities and funding agencies must also determine whether investments in flood mitigation produce sufficient economic returns to justify their costs.

To address this question, a financial scenario analysis was undertaken for Lagos and Durban using the baseline damage estimates developed in the previous chapter. Flood damages were projected annually from **2025 to 2050**, after which alternative mitigation strategies were evaluated using standard investment appraisal techniques.

The analysis compares a **baseline ("Do Nothing")** scenario with selected intervention strategies designed to reduce future flood damages. Each intervention was assessed using Net Present Value (NPV), Benefit-Cost Ratio (BCR) and break-even year.

6.2. Baseline Scenario

The baseline scenario assumes that no major flood mitigation investments are introduced beyond existing infrastructure and maintenance practices.

Under this scenario, flood damages continue to increase over time because of:

- continued urban expansion;
- increasing exposure of people and infrastructure;
- ageing drainage systems;
- increasing rainfall intensity associated with climate change.

The projected analysis covers the period **2025 to 2050** using a **5% annual discount rate**, consistent with many public infrastructure investment studies. These flood risk scores were incorporated into the annual damage projections to reflect the differing levels of vulnerability between the two cities.

Table 7: Baseline scenario parameters used for the flood damage projections, including the projection period, discount rate, and mean flood risk values for Lagos and Durban.

Parameter	**Value**
Projection period	**2025-2050**
Discount rate	**5%**
Lagos mean flood risk	**0.454**
Durban mean flood risk	**0.428**

6.3. Baseline Damage Projections

The projected cumulative flood damages indicate that, without intervention, flood losses continue to increase throughout the study period.

Lagos

Projected cumulative damage reaches approximately **USD 686.4 million** by the end of the projection period.

Durban

Projected cumulative damage reaches approximately **USD 106.3 million** by 2050.

The substantially higher projected losses in Lagos reflect its greater flood susceptibility, larger exposed population and more extensive areas of high-risk urban development.

6.4. Flood Mitigation Scenarios

Different intervention strategies were developed for each city to reflect their contrasting flood mechanisms. Because Lagos experiences widespread urban flooding associated with inadequate drainage, structural drainage improvements were prioritised. Durban's flood behaviour is more strongly influenced by topography and runoff generation, making catchment-based interventions more appropriate. Early warning systems were evaluated for both cities because they represent relatively low-cost measures capable of reducing flood impacts.

6.5. Lagos Intervention Assessment

Two interventions were evaluated.

Drainage Upgrade

This intervention represents large-scale rehabilitation and expansion of the urban stormwater drainage network. The objective is to improve runoff conveyance and reduce surface water accumulation during heavy rainfall. The analysis demonstrates that drainage improvements generate substantial long-term economic benefits. Although the initial investment is large, the intervention prevents more than twice its cost in future flood damages. The positive NPV indicates that the investment creates net economic value over the analysis period.

Table 8: Economic evaluation of the drainage upgrade intervention for Lagos, including investment cost, avoided flood damages, net present value (NPV), benefit-cost ratio (BCR), and break-even year.

Indicator	**Value**
Total investment cost	**USD 1,020 million**
Damage avoided	**USD 2,570.5 million**
Net Present Value	**USD 442.4 million**
Benefit-Cost Ratio	**2.52**
Break-even year	**2041**

Early Warning System

This intervention includes flood forecasting, rainfall monitoring, public alert systems, emergency communication, and evacuation planning. Although the early warning system prevents less total damage than drainage upgrades, it requires substantially lower capital investment. Consequently, it achieves the highest Benefit-Cost Ratio of all interventions evaluated. The investment also reaches financial break-even only six years after implementation, making it particularly attractive for municipalities with limited budgets.

Table 9: Economic evaluation of the early warning system intervention for Lagos, including investment cost, avoided flood damages, net present value (NPV), benefit-cost ratio (BCR), and break-even year.

Indicator	Value
Total investment cost	**USD 288 million**
Damage avoided	**USD 1,165.7 million**
Net Present Value	**USD 374.0 million**
Benefit-Cost Ratio	**4.05**
Break-even year	**2031**

6.6. Comparison of Lagos Interventions

Both interventions produce positive financial returns. However, they differ in important ways. Drainage improvements maximise total flood damage reduction. Early warning systems maximise investment efficiency. The results suggest that early warning systems should be implemented immediately because of their relatively low cost and rapid financial return, while drainage rehabilitation should remain a long-term infrastructure priority.

Table 10: Comparison of the economic performance of the drainage upgrade and early warning system interventions for Lagos.

Measure	Drainage Upgrade	Early Warning
Total Cost	1,020	288
Avoided Damage	2,570.5	1,165.7
NPV	442.4	374.0
Benefit-Cost Ratio	2.52	**4.05**
Break-even Year	2041	**2031**

6.7. Durban Intervention Assessment

Unlike Lagos, Durban requires interventions that reduce runoff generation rather than simply increasing drainage capacity. Two strategies were therefore evaluated.

Retention and Vegetation

This intervention combines detention basins, vegetation restoration, runoff retention, and nature-based flood management. Although the intervention almost recovers its investment costs, the discounted benefits remain slightly below the total investment cost. Consequently, the project does not generate a positive NPV over the study period. Nevertheless, the Benefit-

Cost Ratio of approximately one indicates that the intervention nearly pays for itself financially. It may therefore still be justified when environmental, ecological and social benefits are considered.

Table 11: Economic evaluation of the retention basins and vegetation intervention for Durban, including investment cost, avoided flood damages, net present value (NPV), benefit-cost ratio (BCR), and break-even year.

Indicator	**Value**
Total investment cost	**USD 512 million**
Damage avoided	**USD 522.8 million**
Net Present Value	**−119.5 million**
Benefit-Cost Ratio	**1.02**
Break-even year	**Not achieved**

Early Warning System

The Durban early warning strategy includes rainfall monitoring, river level monitoring, flood forecasting, emergency communication and evacuation planning. Although the financial returns are modest compared with Lagos, the intervention remains economically viable. The positive NPV indicates that benefits slightly exceed costs over the analysis period. Because of the relatively small capital investment, early warning systems represent Durban's most financially attractive intervention.

Table 12: Economic evaluation of the early warning system intervention for Durban, including investment cost, avoided flood damages, net present value (NPV), benefit-cost ratio (BCR), and break-even year.

Indicator	**Value**
Total investment cost	**USD 180 million**
Damage avoided	**USD 244.9 million**
Net Present Value	**USD 1.2 million**
Benefit-Cost Ratio	**1.36**
Break-even year	**2050**

6.8. Comparison of Durban Interventions

Table 13: Comparison of the economic performance of the retention basins with vegetation and early warning system interventions for Durban.

Measure	Retention & Vegetation	Early Warning
Total Cost	512	180
Avoided Damage	522.8	244.9
NPV	−119.5	**1.2**
Benefit-Cost Ratio	1.02	**1.36**
Break-even	Not reached	**2050**

The comparison demonstrates that early warning systems generate better financial performance despite preventing less total damage. Nature-based interventions remain valuable but require either lower implementation costs or longer evaluation periods before their full benefits become apparent.

6.9. Comparison Between Lagos and Durban

The financial analysis reveals clear differences between the two cities. Several factors explain these differences. First, Lagos experiences considerably larger projected flood damages. Reducing these damages therefore produces larger economic savings. Second, the interventions evaluated for Lagos directly address its dominant flood mechanism, namely inadequate drainage. Third, Durban's nature-based solutions generate many environmental and ecological benefits that are not fully captured within conventional financial analysis. Consequently, the financial appraisal likely underestimates their true societal value.

Table 14: Comparison of key economic indicators for flood mitigation interventions in Lagos and Durban, including mean flood risk, highest benefit-cost ratio (BCR), highest net present value (NPV), and earliest break-even year.

Indicator	Lagos	Durban
Mean Flood Risk	**0.454**	**0.428**
Highest BCR	**4.05**	**1.36**
Highest NPV (USD M)	**442.4**	**1.2**
Earliest Break-even	**2031**	**2050**

6.10. Implications for Urban Planning

The scenario analysis demonstrates that investment decisions should be tailored to local flood processes. For Lagos, drainage rehabilitation represents the most effective long-term structural solution. However, early warning systems offer the fastest economic returns and could be implemented immediately while larger infrastructure projects are being planned. For Durban, early warning systems provide the strongest financial performance within the study period. Nature-based interventions remain attractive from a sustainability perspective, particularly when biodiversity conservation, water quality improvement and ecosystem restoration are included in project appraisal.

6.11. Key Findings

The financial analysis produces several important conclusions.

- Flood damages continue to increase under the baseline scenario for both cities.
- Lagos experiences substantially greater projected losses than Durban because of its higher overall flood risk.
- The Lagos drainage programme generates more than **USD 2.57 billion** in avoided flood damages.
- The Lagos early warning system achieves the highest investment efficiency, with a **Benefit-Cost Ratio of 4.05**.
- Durban's early warning system is the only intervention producing a positive Net Present Value within the analysis period.
- Retention and vegetation projects in Durban almost recover their investment costs but require longer planning horizons or broader benefit assessments to demonstrate full economic value.
- Early warning systems consistently provide the greatest return per unit investment and represent an effective first-stage adaptation strategy for both cities.

7. Financial Analysis and Cost-Benefit Assessment

7.1. Introduction

Understanding where flooding occurs is only one part of urban flood management. Decision-makers also need to know whether investing in mitigation measures provides financial returns over time. This section combines the flood risk assessment with economic modelling to evaluate whether selected interventions are economically justified.

The analysis compares a **baseline scenario**, where no additional flood mitigation is implemented, with alternative intervention scenarios for Lagos and Durban. The comparison is based on projected flood damages from **2025 to 2050**, discounted at **5%**, using Net Present Value (NPV), Benefit-Cost Ratio (BCR), total avoided damages, and break-even year.

7.2. Baseline Flood Risk

Spatial modelling indicates that Lagos experiences slightly higher overall flood risk than Durban. Although the average risk values appear similar, Lagos contains more than twice the proportion of land classified as high risk. This reflects extensive low-lying urban development, dense built-up areas, and limited drainage capacity. Durban exhibits lower overall exposure, although steep slopes produce highly localised flash-flood hotspots.

Table 15: Baseline spatial flood risk metrics for Lagos and Durban, showing the mean composite flood risk index and the proportion of land classified as high flood risk.

City	Mean Flood Risk	High-Risk Area (%)
Lagos	0.454	18.9
Durban	0.428	8.5

7.3. Baseline Annual Flood Damage

Regression models were used to estimate present-day annual flood damage. The regression models underestimate historical losses because they rely on broad climatic variables and urban indicators rather than detailed local infrastructure characteristics. This finding reinforces an important conclusion of the study that flood losses cannot be explained by rainfall alone, and that urban vulnerability strongly controls economic damage. Consequently, the scenario modelling uses the historical observations together with projected risk growth rather than relying solely on regression predictions.

Table 16: Comparison of regression-predicted annual flood damages and historically observed average annual flood damages for Lagos and Durban.

City	Predicted Damage (USD Million/year)	Historical Observed Damage (USD Million)
Lagos	15.85	225
Durban	8.54	60

7.4. Baseline Scenario (No Intervention)

Without additional mitigation investment, projected annual flood damage increases steadily between 2025 and 2050. The modelling incorporates continued urbanisation, increasing exposure, projected climate-related increases in flood hazard, and existing infrastructure conditions. The final projected annual damages in 2050 are found in Table 17. Lagos experiences substantially faster growth in economic losses because large portions of the city remain highly exposed to recurrent flooding.

Table 17: Projected annual flood damages under the baseline (no intervention) scenario for Lagos and Durban in 2050.

City	Final Annual Damage (USD Million)
Lagos	686.4
Durban	106.3

7.5. Intervention Scenarios

Separate intervention packages were evaluated for each city.

Lagos

The two interventions that were assessed were drainage upgrades and an early warning system. Drainage upgrades reduce flood occurrence through improved stormwater infrastructure. Early warning systems reduce losses by allowing communities and emergency services to prepare before flood events.

Durban

The two interventions that were assessed were retention basins with vegetation and early warning system. Retention infrastructure reduces runoff and peak discharge, while vegetation improves infiltration and slows overland flow.

7.6. Projected Damage after Intervention

Lagos

Table 18: Projected annual flood damages in 2050 under alternative mitigation scenarios for Lagos.

Scenario	Final Damage in 2050 (USD Million)
No Intervention	686.4
Drainage Upgrade	501.1
Early Warning	604.0

Drainage upgrades produce the largest reduction in future damages, lowering projected annual losses by approximately **185 million USD** by 2050.

Durban

Table 19: Projected annual flood damages in 2050 under alternative mitigation scenarios for Durban.

Scenario	Final Damage in 2050 (USD Million)
No Intervention	106.3
Retention + Vegetation	74.4
Early Warning	92.0

Retention measures provide greater reductions than early warning alone because they directly reduce flood volumes entering vulnerable areas.

7.7. Cost-Benefit Analysis

Four economic indicators were calculated.

1. Total Investment Cost: The total implementation cost over the study period.
2. Avoided Damage: The cumulative reduction in flood losses relative to the baseline.
3. Net Present Value (NPV): The discounted economic benefit after accounting for investment costs. Positive NPV indicates that an intervention creates economic value.
4. Benefit-Cost Ratio (BCR): The ratio of discounted benefits to total costs.

- BCR > 1 indicates benefits exceed costs.
- BCR < 1 indicates costs exceed benefits.

7.8. Economic Evaluation of Flood Mitigation Scenarios

Lagos

Table 20: Cost-benefit analysis results for flood mitigation interventions in Lagos.

Intervention	Cost (USD M)	Avoided Damage (USD M)	NPV (USD M)	BCR	Break-even
Drainage Upgrade	1,020	2,570.5	442.4	2.52	2041
Early Warning	288	1,165.7	374.0	4.05	2031

The drainage programme generates the greatest total reduction in losses. Although it requires a larger capital investment, avoided damages exceed costs by more than a factor of two. The early warning system achieves the highest economic efficiency. Every dollar invested returns approximately four dollars in avoided damages, while recovering its costs approximately ten years earlier than the drainage project.

Durban

Table 21: Cost-benefit analysis results for flood mitigation interventions in Durban.

Intervention	Cost (USD M)	Avoided Damage (USD M)	NPV (USD M)	BCR	Break-even
Retention + Vegetation	512	522.8	-119.5	1.02	Not reached
Early Warning	180	244.9	1.2	1.36	2050

The retention strategy almost recovers its investment costs but does not generate a positive discounted return over the study period. Its negative NPV indicates that, under the assumed costs and discount rate, the investment does not fully pay for itself before 2050. The early warning system produces only modest economic gains, yet it remains financially viable because of its relatively low implementation cost.

7.9. Comparison Between Cities

The financial performance of interventions differs substantially between Lagos and Durban. Lagos shows much stronger economic returns because expected flood damages are considerably higher. Avoiding large losses produces substantial financial benefits that outweigh investment costs. Durban experiences smaller annual damages, reducing the total economic savings achievable through infrastructure investment. The analysis indicates that intervention priorities should differ between the two cities. For Lagos, major drainage improvements provide the greatest reduction in long-term losses, while early warning systems deliver exceptional cost-effectiveness. For Durban, lower-cost measures such as early warning systems offer better financial performance than large capital-intensive infrastructure projects over the 2025 to 2050 analysis period.

7.10. Key Findings

The financial modelling leads to five principal conclusions.

- Flood damages continue to increase under the no-intervention scenario in both cities.
- Lagos exhibits substantially larger long-term economic exposure than Durban.
- Drainage upgrades provide the greatest reduction in total losses for Lagos.
- Early warning systems generate the highest benefit-cost ratio because relatively small investments prevent considerable losses.
- Large infrastructure investments require longer planning horizons before achieving positive economic returns, particularly in Durban.

8. Discussion (Integrated Synthesis of Climate, Regression, Spatial Risk and Financial Results)

This section brings together the climate diagnostics, regression outputs, spatial flood risk surfaces, and financial loss estimates into one interpretation of how flood risk behaves differently in Lagos and Durban. The aim is not to restate results, but to show how the systems connect and where they break standard assumptions about climate-driven flooding.

8.1. Climate signals and the mismatch with flood trends

The climate analysis shows a shared pattern across both cities. Annual rainfall totals are either stable or slightly declining over time. Temperature trends move upward in both Lagos and Durban, with a clearer warming signal in recent decades. At face value, this combination does not suggest a proportional increase in flood frequency or damage.

The flood records show the opposite. Flood events increase over time in both cities, with sharper growth in recent years. This divergence points away from total rainfall as the main driver and towards storm structure and urban exposure. The implication is that rainfall is becoming more concentrated into short-duration, high-intensity events, even when annual totals remain unchanged.

This creates a basic mismatch. The climate signal alone cannot explain the flood signal. Something in the urban system is amplifying exposure and converting rainfall variability into damage.

8.2. Regression results and what they reveal about mechanism

The regression models help separate climate influence from urban structure. For Lagos, precipitation and temperature do not show statistically meaningful relationships with either flood damage or number of people affected. The coefficients are unstable, and model fit remains weak. This suggests that macro climate variables operate too far upstream to explain observed losses. The dominant processes sit elsewhere.

For Durban, precipitation becomes significant in the damage model. This indicates a more direct rainfall response in observed losses. The people affected model shows population as the strongest predictor, with a very high elasticity. Small increases in population are associated with large increases in exposure outcomes. This is consistent with settlement expansion into higher risk terrain and increased density in constrained catchments.

Across both cities, multicollinearity diagnostics show elevated condition numbers. This reflects overlapping explanatory structures, especially between population, urban growth proxies, and climate trends. The models therefore should not be read as clean causal estimators. They instead point to dominant pathways.

The key divergence is structural. Lagos behaves like a system where climate variables lose explanatory power once urban form is considered. Durban retains a partial climate sensitivity but shows strong amplification through population exposure.

8.3. Spatial risk patterns and physical exposure pathways

The spatial flood risk mapping clarifies why the regression behaves this way. Lagos shows a broad, low-lying risk surface with limited topographic variation. Elevation contributes consistently to risk across large continuous areas. Built surface density increases exposure uniformly, producing wide zones of moderate to high risk rather than isolated hotspots. The spatial pattern is horizontal and diffuse. Flooding is less about slope and more about drainage capacity, surface sealing, and water accumulation in flat terrain.

Durban shows a fragmented risk structure driven by elevation and slope. High-risk clusters follow steep valleys, river corridors, and constrained low points. Built density interacts with these features, but does not dominate them. The spatial signal is vertical and segmented, with clear transitions between safe and high exposure zones. This difference explains the regression outcomes. In Lagos, spatial risk is widespread and structurally embedded in land elevation and drainage. Climate variation does not change exposure much because exposure is already saturated. In Durban, spatial risk is conditional. Rainfall events translate into damage when they intersect steep terrain and concentrated settlements.

8.4. Financial loss patterns and exposure scaling

The financial evaluation adds a second layer to the interpretation by translating hazard and exposure into economic outcomes. In Lagos, estimated damages scale more with spatial extent of inundation than with intensity of individual climate variables. Losses accumulate across large areas of moderate risk. This produces a gradual but persistent economic burden rather than highly localised catastrophic losses. In Durban, financial losses are more episodic. High rainfall events in steep catchments produce concentrated damage zones. The cost distribution is uneven, with a smaller number of events generating a disproportionate share of total losses. This aligns with the regression result where precipitation becomes significant, suggesting threshold behaviour in the system.

Population growth plays a stronger role in both cities when viewed through financial exposure, but the mechanism differs. In Lagos, it increases the baseline value at risk across a wide surface. In Durban, it increases the number of assets located in high slope runoff pathways.

8.5. Integrated interpretation across all four components

When the four analytical layers are combined, a consistent structure emerges. Lagos operates as a density-driven flood system on flat terrain. The key constraint is drainage capacity under conditions of rapid urban expansion. Climate variables do not strongly explain variation in damage because the system is already structurally exposed. Flooding is persistent and spatially widespread, with financial losses accumulating through scale rather than intensity. Durban operates as a topography-driven runoff system. Rainfall becomes more important because the landscape converts short-duration events into fast-flowing water movement through constrained valleys. Population growth increases exposure in these high-energy pathways, producing a strong interaction between human settlement and physical terrain.

Both cities show increasing flood impacts despite weak or stable rainfall trends. The difference lies in how water is translated into damage. Lagos translates water into inundation through surface accumulation. Durban translates water into flow through slope and channel concentration.

8.6. Methodological reflection

The results are shaped by data structure as much as physical processes. Annual aggregation reduces the ability to capture short-duration storm effects, which are likely central to flood generation. The use of national-level climate and population proxies introduces spatial smoothing, particularly in Lagos where intra-urban variation is high. The regression models therefore capture broad directional relationships rather than fine-scale causal mechanisms. The spatial models partially correct this by reintroducing terrain structure but still depend on resampling and composite weighting choices. Even with these constraints, the convergence across methods is consistent. Where climate variables weaken, spatial structure strengthens. Where spatial structure is diffuse, economic losses scale with area. Where spatial structure is sharp, losses concentrate in specific catchments.

8.7. Transition to scenario modelling

The combined findings point to different intervention logics for each city. Lagos requires interventions that reduce system-wide exposure under flat terrain conditions. Durban requires interventions that interrupt flow concentration in steep catchments and limit settlement expansion into high-risk corridors. This sets up Step 7, where scenario modelling can test how drainage upgrades, land use controls, and settlement redistribution alter both exposure and expected losses under changing climate intensity patterns.

9. Scenario Modelling and Cost–Benefit Analysis

This section tests how different intervention choices change flood outcomes in Lagos and Durban. The focus shifts from describing risk to altering it. Three scenario types are used: no intervention (baseline), infrastructure improvement, and land-use or exposure control. Each scenario is evaluated through changes in expected flood exposure and simplified cost–benefit comparisons.

9.1. Baseline scenario: continuation of current trajectories

The baseline assumes no structural change in drainage systems, land-use patterns, or settlement growth controls. Climate inputs continue along observed trends: stable to declining annual rainfall with rising temperatures and stronger short-duration rainfall intensity.

Under this scenario:

Lagos shows continued expansion of flood exposure across low-lying zones. Built-up density increases runoff generation faster than drainage capacity can adapt. Flood extent grows laterally, with more frequent shallow inundation across residential and informal settlements.

Durban shows increasing concentration of flood events in steep catchments. The spatial footprint of flooding remains narrower, but peak intensity increases. Damage is episodic, driven by rainfall events that exceed channel capacity in confined valleys.

Economically, Lagos produces high cumulative losses due to repeated moderate events. Durban produces lower frequency but higher variance losses, with occasional high-cost disasters.

9.2. Infrastructure upgrade scenario

This scenario models improvements in drainage capacity, stormwater systems, and channel conveyance. It assumes partial reduction in flood depth and duration, without changing urban form.

Lagos: drainage expansion and pumping efficiency

In a flat terrain system, infrastructure upgrades primarily affect water residence time. Improved drainage reduces inundation duration rather than eliminating flood extent. Expected outcomes were reduced frequency of low-to-moderate flood losses, limited effect on extreme events during high-intensity rainfall and persistent risk in highly sealed zones with limited infiltration capacity. The cost-benefit profile is sensitive to maintenance. Benefits decline sharply when systems are undersized relative to rapid urban growth.

Durban: channel and culvert upgrades

In Durban, infrastructure upgrades directly interact with slope-driven flow. Enlarged culverts and reinforced channels reduce peak discharge bottlenecks. Expected outcomes were a noticeable reduction in event-based damage in known hotspots, partial decoupling of rainfall intensity from damage outcomes, and residual risk remains in informal settlements on steep slopes where relocation is absent. The benefit is higher per unit investment compared to Lagos because upgrades target spatially concentrated flow pathways.

9.3. Land-use and exposure control scenario

This scenario focuses on reducing exposure rather than modifying hazard. It includes zoning enforcement, relocation from high-risk areas, and restrictions on new development in flood-prone zones.

Lagos: densification control in low-lying areas

Lagos benefits strongly from exposure management because risk is widespread but shallow. Preventing additional settlement in flood-prone coastal and lagoon-adjacent zones reduces the exposed population footprint. Expected outcomes were a large reduction in cumulative economic losses over time, slower growth in affected population rather than immediate risk elimination and strong dependence on enforcement capacity. This scenario produces compounding benefits because avoided exposure accumulates over time.

Durban: slope-based settlement restriction

In Durban, exposure control targets high-risk terrain corridors. Preventing settlement expansion into steep valleys and riverbanks directly reduces high-impact event potential. Expected outcomes were a significant reduction in catastrophic loss probability, lower variability in annual damages and persistent exposure in existing informal settlements unless relocation is included. Exposure control in Durban has higher marginal impact because it targets high-energy flow zones rather than diffuse floodplains.

9.4. Combined scenario: infrastructure plus exposure control

The combined intervention produces the most stable outcomes in both cities, but through different mechanisms. In Lagos, drainage improvements reduce routine flooding while exposure control limits long-term expansion into vulnerable low-lying areas. Together, they shift the system from chronic inundation toward managed drainage stress. In Durban, infrastructure upgrades reduce peak flow impacts while land-use controls prevent new settlements in high-risk catchments. This combination reduces both frequency and severity of flood damage. The interaction effect is important. Infrastructure alone does not stop exposure growth. Exposure control alone does not manage existing flood pathways. Together, they reshape both sides of the risk equation.

9.5. Cost-benefit interpretation

A simplified comparison of costs and avoided losses shows different efficiencies across cities.

In Lagos:

- High upfront infrastructure costs due to scale and maintenance needs
- Moderate reduction in per-event losses
- Strong long-term returns only when paired with exposure control

- Benefit curve flattens under continued urban expansion

In Durban:

- Moderate infrastructure costs focused on targeted catchments
- High reduction in high-loss events
- Strong return on investment when high-risk zones are prioritised
- Exposure control produces immediate reduction in tail risk

The key difference is efficiency of intervention. Lagos requires system-wide adjustments. Durban allows targeted interventions with clearer spatial payoff.

9.6. Scenario comparison across analytical layers

When mapped back onto earlier findings:

- Climate trends remain unchanged across scenarios
- Regression relationships shift indirectly through reduced exposure
- Spatial risk surfaces are the primary driver of scenario effectiveness
- Financial losses respond non-linearly to exposure reduction, especially in Durban

This confirms that flood outcomes in both cities are more sensitive to spatial organisation than to climate variation alone.

9.7. Final synthesis for policy direction

The scenario results reinforce a split adaptation logic. Lagos benefits most from reducing how much of the city occupies flood-prone flat terrain while upgrading drainage to manage persistent surface water accumulation. Durban benefits most from limiting settlement in steep catchments and reinforcing a smaller number of high-risk hydraulic corridors. Both systems respond more strongly to exposure management than to climate correction. Climate sets the pressure. Urban form determines the damage.

10. Conclusion and Research Implications

This section closes the analysis by pulling together the climate trends, regression outputs, spatial patterns, financial results, and scenario testing into a single interpretation of how flood risk operates in Lagos and Durban. It also sets out what the findings mean for urban flood research and applied planning.

10.1. Core findings across the full pipeline

Across all four analytical layers, one pattern repeats. Climate trends alone do not track flood outcomes. Annual rainfall is stable or slightly declining in both cities, while flood events increase. Regression models show weak explanatory power for macro climate variables in

Lagos, and mixed but partial significance in Durban. Population becomes a stronger driver of exposure than climate in both cases, especially for people affected.

Spatial analysis shows two distinct physical systems:

- Lagos: flat terrain, diffuse flooding, drainage-limited system
- Durban: steep terrain, channelised flow, slope-controlled system

Financial losses follow these structures. Lagos accumulates cost through repeated moderate events across wide areas. Durban concentrates cost in fewer but sharper events tied to specific catchments. Scenario modelling confirms that changing exposure produces larger gains than adjusting climate sensitivity alone.

10.2. Integrated interpretation of flood mechanisms

The results point to two different flood mechanisms operating under similar climate signals. In Lagos, flooding behaves as a surface saturation problem. Water spreads slowly across low-lying built environments where drainage and infiltration are limited. Risk expands outward as urban density increases. Climate variability matters less because the system is already structurally prone to ponding. In Durban, flooding behaves as a flow concentration problem. Rainfall is rapidly converted into runoff through steep gradients. Water accelerates through valleys and constrained channels. Damage depends on where people and infrastructure sit along these flow paths. Both systems respond to rainfall extremes, but the pathway from rainfall to damage is different.

10.3. Answering the research questions

Why is flooding happening in both cities?

Flooding is driven primarily by urban form interacting with local geography. In Lagos, sealed surfaces and low elevation dominate. In Durban, slope and catchment structure dominate. Climate trends act as a trigger rather than a sole cause.

Has flooding increased over time?

Yes. Flood frequency increases in both cities despite stable or declining annual rainfall. This aligns with increasing urban exposure and more intense short-duration rainfall events.

How far does flood damage reach spatially and economically?

Lagos shows wide spatial spread with moderate depth of impact. Economic losses accumulate across large areas. Durban shows concentrated spatial hotspots with higher intensity damage and sharper economic peaks.

What could reduce future flood risk?

Exposure control consistently produces the strongest reduction in losses. Infrastructure upgrades improve outcomes but depend on spatial targeting. The most effective pathway combines land-use regulation with drainage or hydraulic system upgrades.

10.4. Methodological reflection

The workflow produces consistent results across different data types, but each layer carries constraints. Annual aggregation reduces sensitivity to storm-level extremes. National-level proxies smooth city-scale variation. Resampling in spatial analysis introduces uncertainty in edge zones. Regression models are affected by multicollinearity between population and urban expansion indicators. Even with these limitations, the convergence across methods is stable. Where one method weakens, another reinforces the same structural interpretation.

10.5. Contribution of the study

The main contribution is the separation of climate signal from urban structure in explaining flood outcomes. Instead of treating flooding as a direct response to rainfall trends, the analysis shows that climate sets boundary conditions, urban form determines exposure, topography controls flow pathways, and population growth scales losses. This shifts the explanation from a climate-centred view to a system interaction view.

10.6. Closing synthesis

Lagos and Durban do not behave like variations of the same flood system. They represent two different physical–urban configurations under similar climate pressure. One spreads water across a flat, densely built surface. The other channels water through steep, constrained terrain. The difference between them is not the amount of rainfall received. It is how the city structure turns rainfall into damage.

11. Limitations and Future Research Directions

This section outlines where the analysis holds, where it bends, and what would need to change for sharper inference. It also points to extensions that could turn this work into a more granular decision tool for flood planning.

11.1. Data constraints and measurement limits

The analysis relies on multiple global and national datasets stitched into city-level proxies. This creates consistency across variables but also introduces smoothing.

Climate data

ERA5 provides reliable long-term climate signals, yet it operates at a resolution that can miss short, localised convective storms. Those are often the events that drive urban flooding in both Lagos and Durban. Annual aggregation further flattens extremes. A single high-intensity storm becomes indistinguishable from multiple moderate events.

Flood records

EM-DAT captures major disaster events, but smaller recurring floods are underrepresented. This biases the dataset toward high-impact episodes and reduces sensitivity to chronic flooding in informal settlements.

Socioeconomic variables

Population and exposure indicators are derived from national or coarse urban estimates. In Lagos especially, intra-city variation in density and settlement type is large enough to shift local flood risk significantly.

11.2. Spatial modelling limitations

The spatial risk model is useful for comparison, but it carries structural assumptions. Resampling DEM and built surface layers to a common resolution introduces interpolation error. Sharp transitions in terrain or settlement density can be softened. The weighting scheme (elevation 40%, slope 30%, built density 30%) reflects a defensible starting point, but it is still subjective. Different weightings would shift hotspot intensity, especially in transitional zones. Drainage infrastructure is not explicitly modelled. This is a major omission in Lagos, where engineered systems strongly influence flood behaviour, and in Durban where culverts and channel capacity control peak flow impacts.

11.3. Statistical modelling limitations

The regression framework identifies associations rather than causal mechanisms. Multicollinearity between population, urban expansion, and climate trends reduces interpretability of coefficients. Condition numbers confirm instability in some specifications. Low observation counts after merging datasets limit statistical power. This is especially relevant for detecting climate effects that operate over longer lags or non-linear thresholds. Log transformations improve distributional behaviour but also compress extreme values, which are often the most policy-relevant outcomes.

11.4. Structural limitations in the analytical design

The study separates climate, spatial, and financial systems into distinct modules. This improves clarity, but flood systems are coupled.

In reality:

- rainfall intensity affects runoff generation
- runoff interacts with drainage capacity
- drainage performance depends on maintenance and governance
- exposure evolves simultaneously through urban growth

These feedbacks are not fully captured in the current framework.

11.5. Directions for improved modelling

Several extensions would strengthen the analytical resolution.

1. **Event-based flood modelling**

Moving from annual averages to storm-event datasets would allow direct linkage between rainfall intensity, runoff response, and damage outcomes. This would reduce the loss of peak-event information.

2. **Higher resolution urban datasets**

Using GHSL or WorldPop at finer spatial scales could improve representation of informal settlements and heterogeneous density patterns, especially in Lagos.

3. **Hydrodynamic or runoff simulation**

Coupling the spatial risk model with a simplified hydrological model (e.g. runoff accumulation or flow direction modelling) would allow explicit simulation of water movement rather than static risk scoring.

4. **Non-linear statistical models**

Generalised additive models or machine learning approaches could capture threshold effects observed in Durban, where damage escalates rapidly after rainfall intensity limits are crossed.

5. **Drainage infrastructure layer**

Digitising stormwater networks and channel capacity would close a major gap in both cities. This is likely to change spatial risk rankings significantly.

11.6. Comparative research extensions

The Lagos-Durban comparison opens space for broader replication. Extending the framework to other coastal African cities such as Accra, Dar es Salaam, or Maputo would test whether the same two-system pattern holds:

- flat, drainage-limited urban flooding systems
- slope-driven runoff systems

This would help determine whether the observed structures are generalisable across coastal African urbanisation patterns or specific to the two case studies.

11.7. Final methodological note

The strongest result of the analysis is not a single coefficient or map. It is the alignment between independent methods. Climate trends, regression outputs, spatial risk, and financial losses all point in the same direction: flooding intensity is less about total rainfall and more about how cities are built into their landscapes. The uncertainty lies in magnitude and precision, not in the direction of the relationship.

12. Conclusion

This study aims to uncover the reasons behind the increasing flooding in Lagos and Durban, exploring how it changes over time, how it spreads across different areas, and what interventions can alter these outcomes.

Through climate analysis, regression modelling, spatial risk mapping, and financial assessments, a consistent pattern emerges. Interestingly, flooding does not correlate well with annual rainfall trends; both cities exhibit stable or even declining precipitation totals, yet flood events continue to rise. While rising temperatures and intensified short-duration rainfall patterns align more closely with this phenomenon, they fail to fully explain the observed damage.

The key factor lies in urban design. Lagos, with its low-lying, high-density environment, tends to trap water, causing it to accumulate more quickly than it can drain away. Flooding becomes widespread and lingers because of minimal elevation variation and drainage systems that are unable to keep pace with urban growth. Though climate variability plays a role, it becomes significant only in an already saturated surface environment.

Durban presents a different scenario. In this city, steep slopes and narrow valleys shape how rainfall transitions into runoff. Water flows swiftly along defined routes, leading to concentrated flood zones. The level of damage in Durban largely depends on where settlements are situated along these drainage paths. As populations move into these vulnerable areas, the risk of flooding increases, making rainfall intensity closely tied to loss outcomes compared to Lagos.

Spatial modelling highlights these differences. In Lagos, flood risk is more diffuse, affecting broader areas, whereas Durban shows concentrated hotspots influenced by terrain. The financial implications follow a similar pattern, with Lagos experiencing cumulative moderate losses and Durban facing sporadic high-impact damages.

Scenario testing indicates that managing exposure is the most effective strategy for reducing future losses in both cities. Although improving infrastructure can lessen damage, its success hinges on strategic spatial targeting and ongoing maintenance. Consistently, land-use regulation proves to be a more potent tool, as it changes how and where risks can accumulate.

Across all methods applied in this analysis, one clear relationship emerges: while climate establishes the conditions that lead to flooding, the urban landscape dictates the extent of the damage.

This study does not rely on a single dataset or methodology; rather, it reaches the same conclusion through various climate trends, statistical models, spatial analyses, and cost estimates. Each analytical layer sheds light on different aspects of the system, yet they all converge on the same underlying mechanism. Ultimately, flood risk in Lagos and Durban is shaped less by the amount of rain that falls and more by how each city interacts with its landscape.